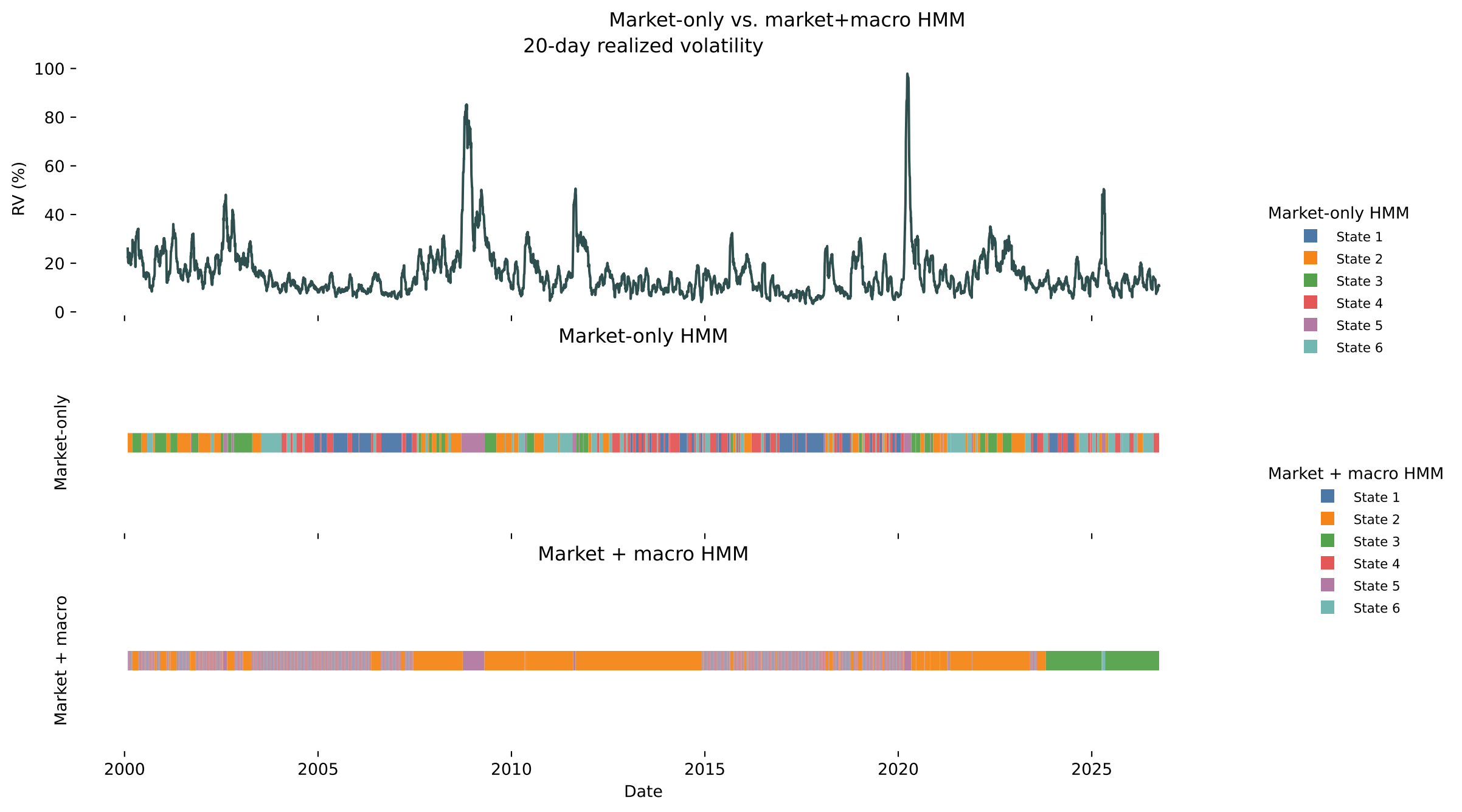

Market-only vs. market+macro HMM
20-day realized volatility
RV (%)
100
80
60
40
20
0
Market-only HMM
Market-only
Market + macro HMM
Market + macro
2000
2005
2010
2015
2020
2025
Date
Market-only HMM
State 1
State 2
State 3
State 4
State 5
State 6
Market + macro HMM
State 1
State 2
State 3
State 4
State 5
State 6